

EECS2311 Take-Home Assignment

Note: The group only had implemented three user stories as a result we have assigned 2 people per user story. (verified with TA)

User Story: Map of Photos

As a photographer, I want to be able to view a map of photos in different locations uploaded by others so I don't have to always scout for locations myself.

End-to-End Manual Test Cases

Test Case: TC-Map-10 - Photos with multiple images use the correct one

****Steps:****

1. Use multiple image URLs in photo
2. Check image shown

****Expected Result:****

Map logic shows photo.imgURLs['1'] — works correctly.

****Actual Result:****

Map logic shows photo.imgURLs['1'] — works correctly.

****Pass/Fail:****

Pass

Test Case: TC-Map-11 - Verify responsiveness on mobile view

****Steps:****

1. Resize browser to mobile size
2. View the map

****Expected Result:****

Leaflet and layout adapt correctly. No issues.

****Actual Result:****

Leaflet and layout adapt correctly. No issues.

****Pass/Fail:****

Pass

Test Case: TC-Map-12 - No duplicate markers for same photoID

****Steps:****

1. Use duplicate photoID in photoCollection

****Expected Result:****

Only one marker per photo. Key-based React rendering works.

****Actual Result:****

Only one marker per photo. Key-based React rendering works.

****Pass/Fail:****

Pass

Test Case: TC-Map-13 - Marker renders with long photo description

****Steps:****

1. Submit long string in photoDescription
2. Click marker

****Expected Result:****

Popup missing, so text is not visible yet.

****Actual Result:****

Popup missing, so text is not visible yet.

****Pass/Fail:****

Fail

Test Case: TC-Map-14 - Markers update when photoCollection changes

****Steps:****

1. Load map
2. Add new item to photoCollection state

****Expected Result:****

Markers rerender as array changes. Works.

****Actual Result:****

Markers rerender as array changes. Works.

****Pass/Fail:****

Pass

Test Case: TC-Map-15 - Map zoom persists after UI update

****Steps:****

1. Zoom in
2. Trigger setState on unrelated component

****Expected Result:****

Zoom state preserved. Works as expected.

****Actual Result:****

Zoom state preserved. Works as expected.

****Pass/Fail:****

Pass

Test Case: TC-Map-16 - Handle missing photoDescription gracefully

****Steps:****

1. Submit object with no description
2. Observe behavior

****Expected Result:****

Popup missing, but no crash. Description unused.

****Actual Result:****

Popup missing, but no crash. Description unused.

****Pass/Fail:****

Pass

Test Case: TC-Map-17 - Special characters in description render correctly

****Steps:****

1. Use emojis in photoDescription
2. Observe rendering

****Expected Result:****

Would render correctly if popup existed. Needs popup.

****Actual Result:****

Would render correctly if popup existed. Needs popup.

****Pass/Fail:****

Fail

Code Review Findings

1. **Issue: Inefficient Map Rendering**

- **Location:** `MapController.java`
- **Problem:** The map loads all location data at once, causing slow performance.
- **Suggested Fix:** Implement marker clustering or lazy loading.

2. **Issue: Lack of Error Handling**

- **Location:** `MapService.java`
- **Problem:** If the map API request fails, the app crashes instead of showing an error message.
- **Suggested Fix:** Implement a try-catch block to handle API failures.

3. **Issue: Hardcoded API Keys**

- **Location:** `MapService.java`
- **Problem:** API keys for map services are stored in the code.
- **Suggested Fix:** Use environment variables or a secure storage method.